

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

8875/01

Paper 1 Multiple Choice

24 September 2014

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

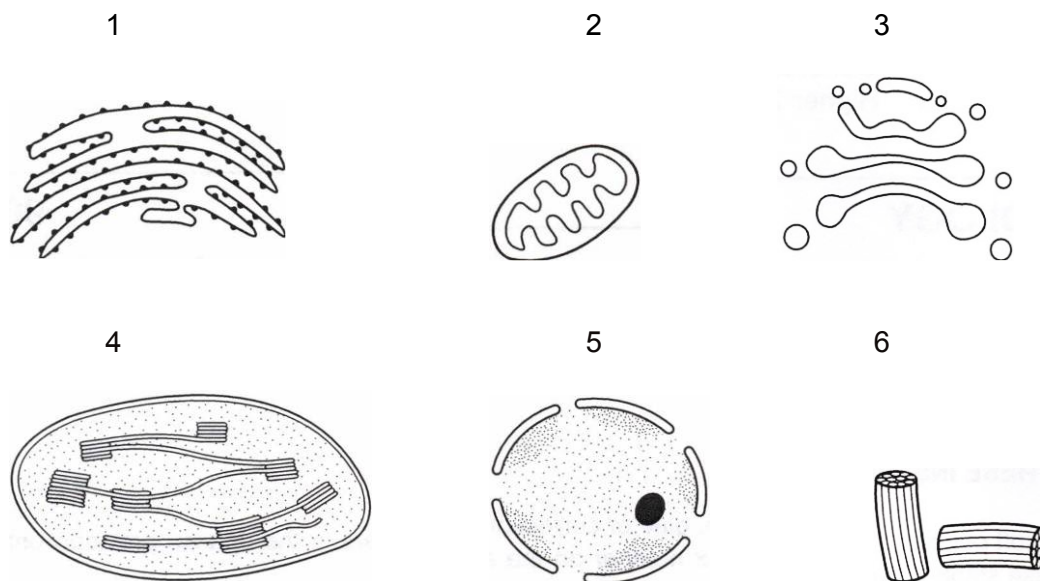
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Calculators may be used.

This document consists of **13** printed pages and **1** blank page.



1 The diagrams are of eukaryotic cellular organelles.



Which row identifies the typical distribution of these organelles?

	animal only	plant only	animal and plant
A	1, 3, 6	2	4, 5
B	3, 5	4	1, 2, 6
C	4, 6	1, 5	2, 3
D	6	4	1, 2, 3, 5

2 Which molecules, found in cell surface membranes, contribute to cell recognition?

- A** Cholesterol and glycolipids
- B** Cholesterol and phospholipids
- C** Glycolipids and glycoproteins
- D** Phospholipids and glycoproteins

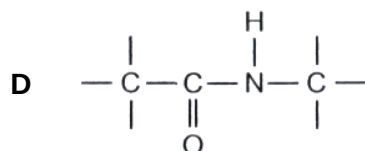
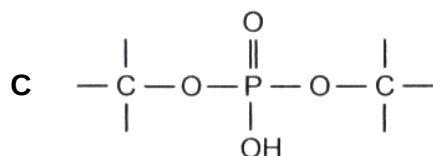
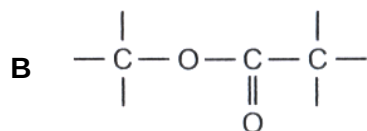
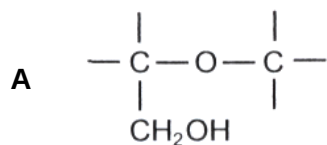
3 Compared to globular proteins, fibrous proteins are

- A** less regular in structure.
- B** less stable chemically.
- C** more readily soluble.
- D** more resistant to high temperatures.

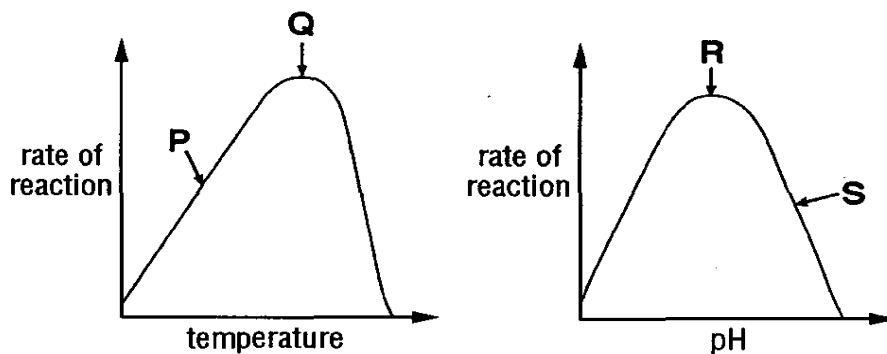
4 Which feature is true of cellulose and accounts for its strength?

- A** It contains both 1, 4 and 1, 6 links between glucose units.
- B** It has OH groups that form hydrogen bonds with adjacent cellulose molecules.
- C** It is a fibrous molecule which is insoluble in water.
- D** It is made up of long, branched and coiled chains of β -glucose units with 1, 4 links.

5 Which set of bonds is found in triglycerides?



6 The graphs show the effects of temperature and pH on enzyme activity.



Which statement is a correct explanation of the rate of reaction at the point shown?

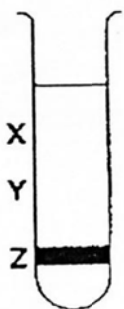
- A At P, hydrogen bonds in the enzyme are broken.
- B At Q, the kinetic energy of enzyme and substrate is highest.
- C At R, covalent bonds are formed between enzyme and substrate.
- D At S, ionic bonds in the enzyme are broken.

- 7 Which statement is true of all enzymes?
- A They are denatured at temperatures above 60°C.
 - B They are inhibited at low pH values.
 - C They catalyse the breakdown of large molecules into smaller ones.
 - D They reduce the amount of energy required to start a reaction.
- 8 Which pair(s) of factors is inversely proportional to the rate of diffusion?
- 1 Concentration gradient and surface area over which diffusion occurs.
 - 2 Distance over which diffusion occurs and size of diffusing molecule.
 - 3 Size of diffusing molecule and concentration gradient.
 - 4 Surface area over which diffusion occurs and distance over which diffusion occurs.
- A 2 only
 - B 1, 2 and 4 only
 - C 2, 3 and 4 only
 - D 1, 2, 3 and 4
- 9 Which is the shortest phase in the mitotic cell cycle?
- A anaphase
 - B interphase
 - C prophase
 - D telophase
- 10 The table shows the results of an analysis of percentage concentration of three bases in nucleic acids from four sources. Three of the sources are DNA and one is RNA. Which source is RNA?

source	adenine	cytosine	guanine
A	18.9	31.9	30.2
B	25.5	24.6	23.8
C	18.8	26.7	31.4
D	30.4	18.5	18.7

- 11** A culture of bacteria had all its DNA labelled with the heavy isotope of nitrogen, ^{15}N . The culture was then allowed to reproduce using nucleotides containing normal ^{14}N . The DNA was examined using a centrifuge after one generation and again after two generations.

The diagram shows the position of the DNA band at Z in the centrifuge tube when the DNA was first labelled.



In which pattern would the DNA be found after the first and after the second cell generations?

	After first generation	After second generation
A	Half at X and half at Y	Quarter at X and at Z and half at Y
B	Half at X and half at Z	Quarter at X and at Z and half at Y
C	All at X	Half at X and half at Y
D	All at Y	Half at X and half at Y

- 12** Which of the following statements are true of the end-replication problem of linear DNA?

- 1 When a linear DNA molecule replicates, a gap is left at the 3' end of each new strand because DNA polymerase can only add nucleotides to a 5' end.
- 2 Telomerase prevents the end-replication problem from occurring.
- 3 Repeated rounds of replication produce shorter and shorter DNA molecules.
- 4 Prokaryotes do not have the end-replication problem.

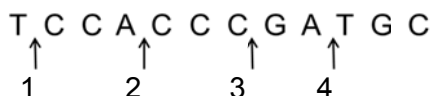
- A** 1 and 2
B 1 and 4
C 2 and 3
D 3 and 4

- 13** If there were 35 amino acids and DNA only contained three types of nitrogenous bases, what would be the minimum number of bases per codon that could code for proteins?

- A** 3
B 4
C 6
D 8

- 14** The RNA triplet UAG acts as a stop codon which terminates the synthesis of a polypeptide. The diagram shows a strand of DNA which codes for four amino acids.

In which position will an insertion of a thymine nucleotide result in the termination of translation?



- A** 1
B 2
C 3
D 4
- 15** Which of the following statements are correct?
- 1 A mutation in an allele occupying a gene locus may result in a change in the individual's phenotype.
 - 2 Gene mutations always give rise to recessive alleles.
 - 3 Chromosomal mutations can give rise to new alleles.
 - 4 Aneuployploids have extra chromosomes in their genome.
- A** 1 only
B 1 and 4
C 1, 3 and 4
D 2, 3 and 4
- 16** In a species of plant, the allele for tall stems is dominant over that for short stems and the allele for purple flowers is dominant over that for white flowers.

A cross between two plants produced the following phenotypic ratio:

tall, purple-flowered	3
tall, white-flowered	3
short, purple-flowered	1
short, white-flowered	1

What are the phenotypes of the parents?

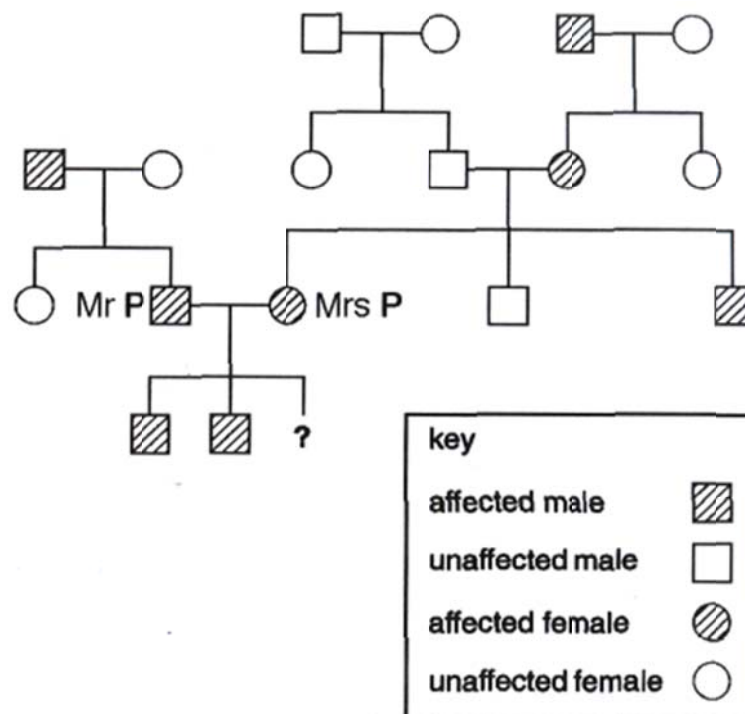
- A** short, purple-flowered x tall, purple-flowered
B short, white-flowered x tall, purple-flowered
C tall, purple-flowered x tall, white-flowered
D tall, white-flowered x short, white-flowered

- 17 In Shorthorn cattle, the allele for hornless is dominant to the allele for the presence of horns. Coat colour can be red (genotype $C^R C^R$), roan (genotype $C^R C^W$) or white (genotype $C^W C^W$). A roan bull, heterozygous for the hornless trait, is crossed with a cow of the same genotype.

What is the probability that a calf from this cross would have the same genotype as its parents?

- A 1/4
- B 3/8
- C 1/2
- D 3/4

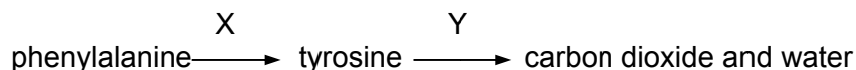
- 18 The family tree was assembled by a genetic counsellor for Mr and Mrs P who suffer from a rare heart disease. Children who inherit the dominant mutant allele from both parents rarely survive beyond puberty.



What is the probability that Mr and Mrs P's third child will be unaffected?

- A 0.75
- B 0.50
- C 0.25
- D 0.00

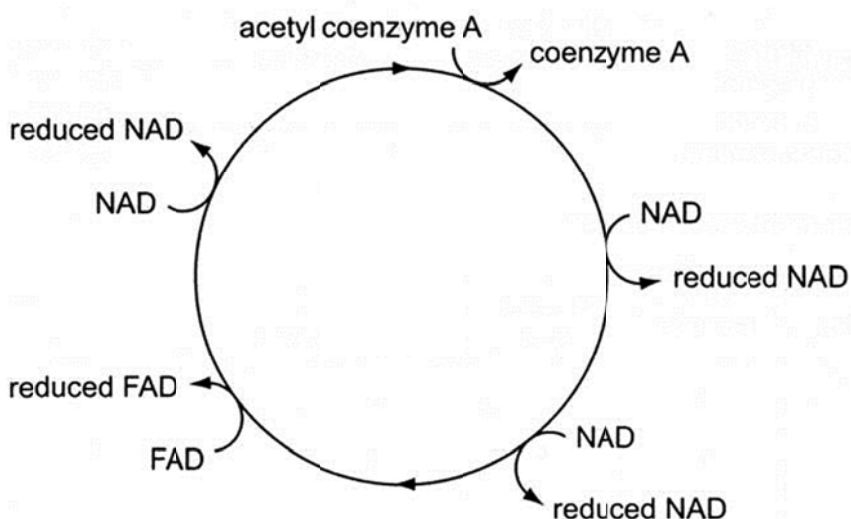
- 19 The following reaction sequence occurs in humans.



Genetic disease P is caused by an enzyme deficiency in step X and genetic disease Q is caused by an enzyme deficiency in step Y. Both conditions are rare and are caused by recessive alleles.

A person with genetic disease P marries a person with genetic disease Q. Which phenotypes would be expected for their children?

- A all have neither genetic disease
 - B all have genetic disease P only
 - C all have genetic disease Q only
 - D all have both genetic diseases
- 20 The diagram shows the reactions of the hydrogen carriers in the Krebs cycle.

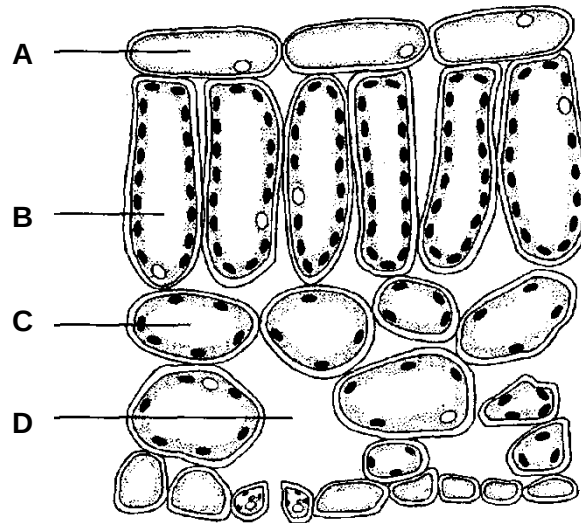


The average yield of ATP, in oxidative phosphorylation, is 2.5 molecules from each molecule of reduced NAD and 1.5 molecules from each molecule of reduced FAD.

What is the average yield of ATP from the hydrogen carriers reduced in the Krebs cycle from one molecule of glucose?

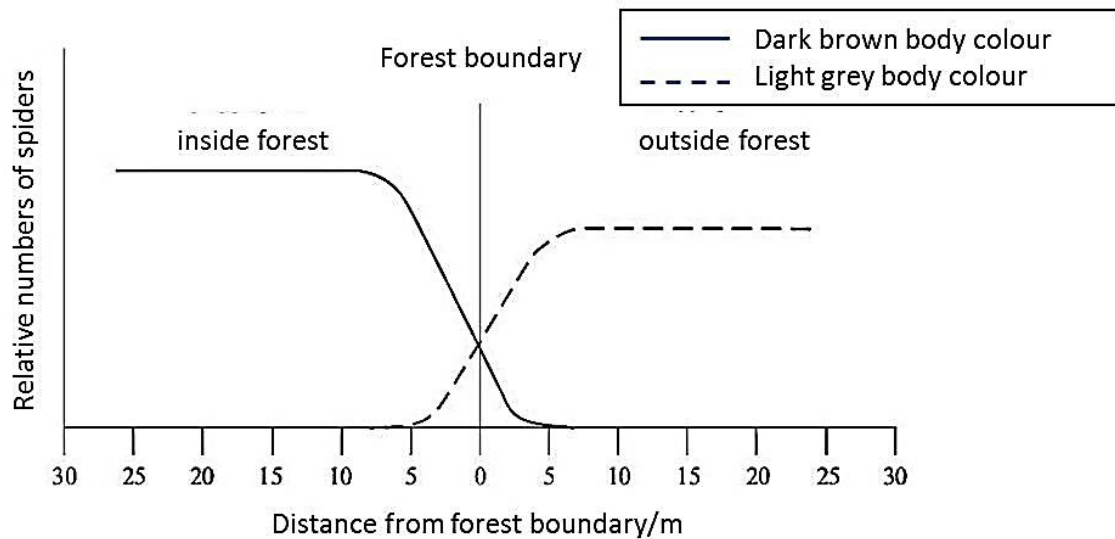
- A 9
- B 18
- C 28
- D 32

- 21 The diagram represents a transverse section of a leaf. Which region is the main site of carbon fixation?



- 22 A particular type of autoimmune disease causes destruction acetylcholine receptors of neurons. What effect will this have on the functioning of the nervous system?
- A Depolarized neurons will be unable to restore resting potential.
 - B Neurons will be unable to propagate action potentials along their axons.
 - C Presynaptic neurons will be unable to synthesize acetylcholine.
 - D Postsynaptic neurons will be unable to detect signals from presynaptic neurons.
- 23 Many types of evidence, including homology, can provide evidence in support of Darwin's theory of natural selection. Which statement does not provide support?
- A The allele for sickle cell haemoglobin that gives resistance to malaria is more frequent in malarial areas.
 - B The distribution of the variants of the A blood group antigen reflects human migration patterns.
 - C The homozygous condition of the sex-linked allele for a non-functional blood clotting protein is rare.
 - D The molecular structure of ATP is almost identical in all eukaryotes.

24 The graph below shows the distribution of huntsman spiders at a forest boundary.



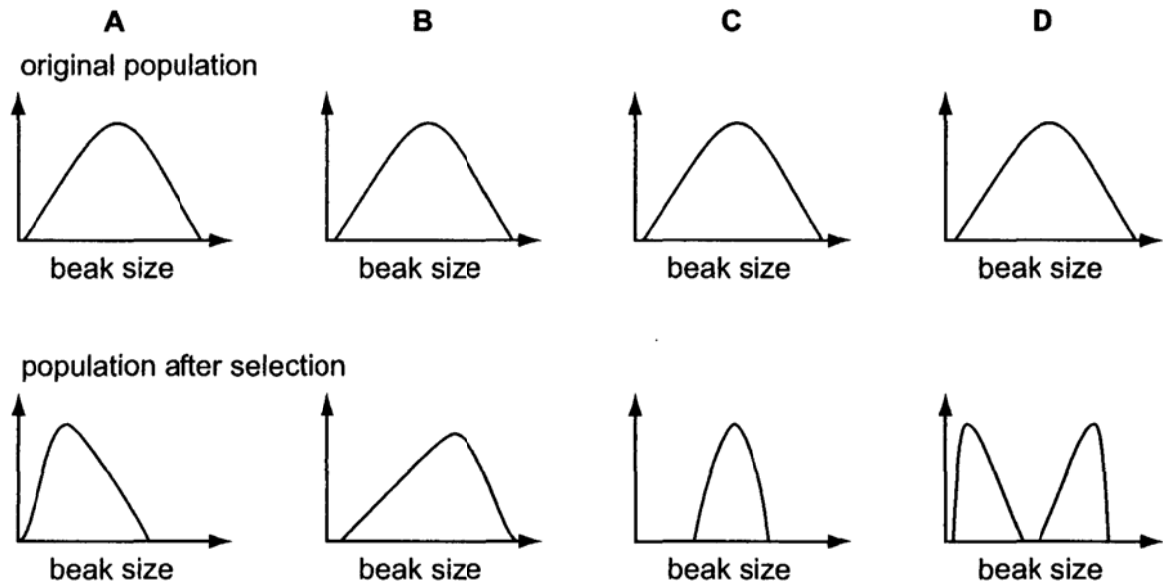
One species of huntsman spider (*Isopeda isopedella*) varies in body colour from dark brown to light grey. In one community at the forest boundary, two populations of this species were found. Some were found living in the leaf litter inside the forest and others were found living in the grass just outside the forest. The relative numbers of dark brown adult spiders and light grey adult spiders found at certain distances from the forest boundary are shown in the graph above.

The best explanation for this distribution is that the

- A differences in the two habitats had changed the physical appearance of individual spiders.
- B particular body colour provided a selective advantage to spiders in a particular habitat.
- C two populations of spiders were once different species.
- D two populations of spiders were unable to interbreed and individuals were adapting to suit their habitats.

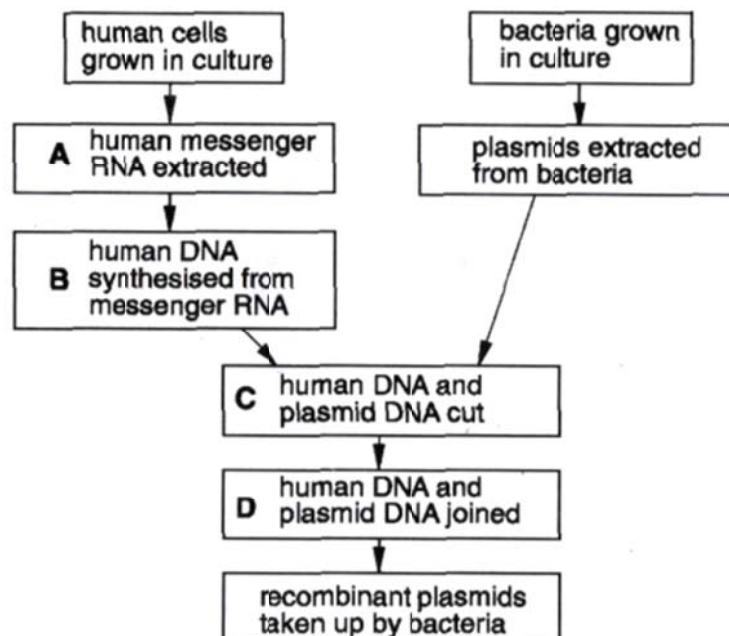
- 25 One species of finch on an island shows variation in beak size. Birds with larger beaks can eat larger seeds. Following a dry period, fewer small seeds are available, with larger seeds and nuts being more plentiful.

Which graph shows the effect of natural selection on beak size of these birds following a dry period?



- 26 The flow diagram shows some of the events in the production of a human hormone by genetic engineering.

At which stage in the process is a restriction enzyme used?

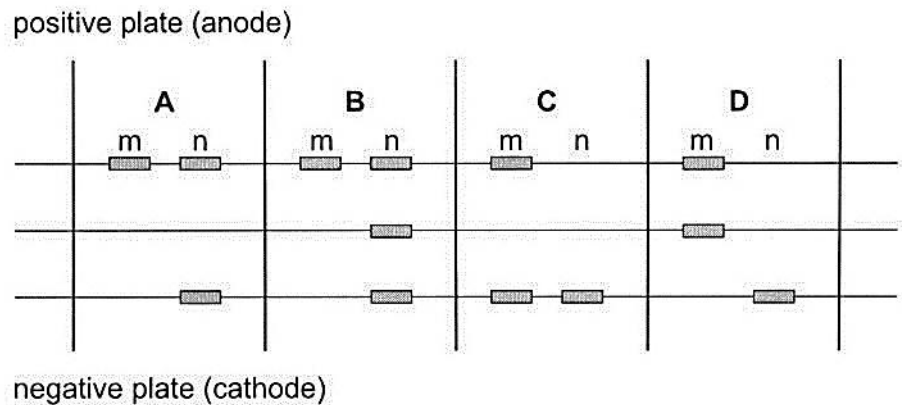


- 27 There are two forms of the seeds of the garden pea, *Pisum sativum*, one with a wrinkled skin and the other with a smooth skin.

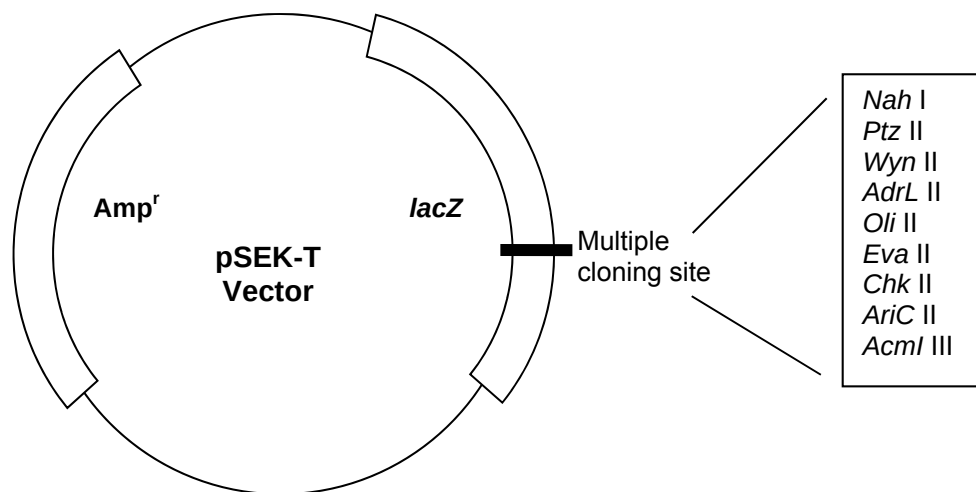
Plants with wrinkled seeds have a recessive mutation caused by the insertion of a transposable element into a gene.

Two heterozygous pea plants were crossed. Two of the offspring (m and n) had the DNA for this gene removed and separated by electrophoresis.

Which pattern of bands shows a wrinkled phenotype and a smooth phenotype?



- 28 To produce recombinant proteins in *E. coli*, the gene encoding for the Bt toxin was inserted into the pSEK-T vector. The restriction sites and selectable markers on the vector are shown below.

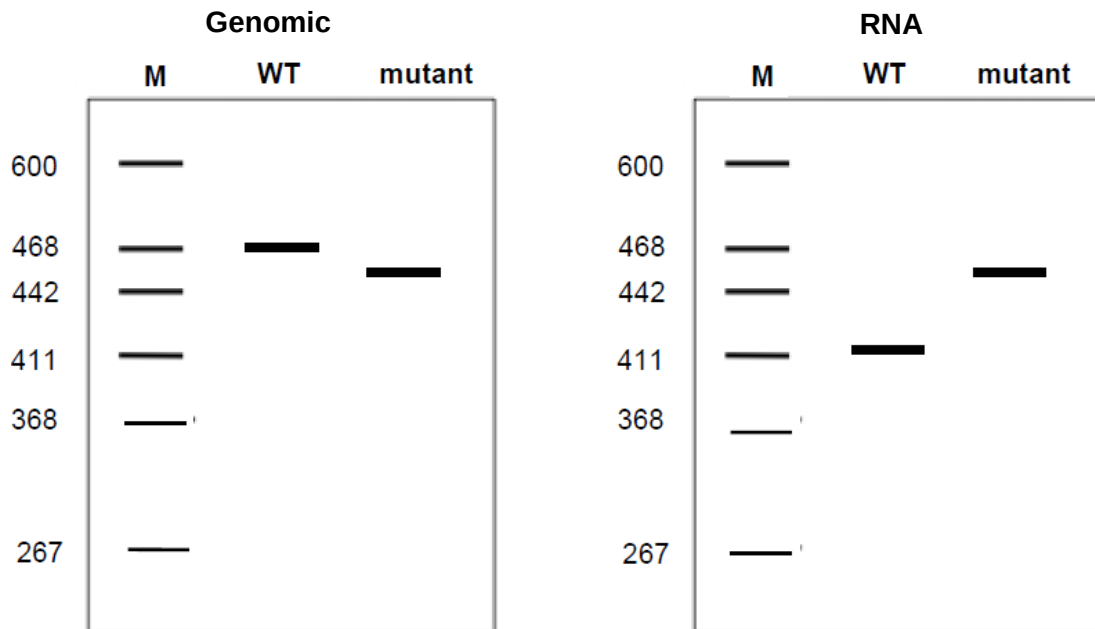


Which of the following chemicals, when added to nutrient medium, would allow screening of recombinant bacteria if the gene for Bt toxin were to be inserted into the multiple cloning site?

	chemicals		colour of colonies
A	ampicillin	X-gal	blue
B	ampicillin	X-gal	white
C	β -galactosidase	X-gal	blue
D	β -galactosidase	lactose	white

- 29 In separate reactions, PCR was performed using a pair of primers with sequences complementary to parts of separate exons of a gene suspected to be involved in a genetic disease.

Using genomic DNA and RNA isolated from healthy (wild-type) and diseased cells, separate PCR reactions were done. The PCR products were analysed on polyacrylamide gels. The sizes of the bands in the molecular weight marker (M) are indicated in base-pairs.



Which of the following **best** explains the results obtained?

- A Deletion of an exon
 - B Deletion of several introns
 - C Deletion of several codons
 - D Deletion of a splice site
- 30 Which of the following genetic modification organism could increase the yield measured as mass of produced per unit area per year?
- A Bt corn
 - B Glyphosate resistant plants
 - C GM salmon
 - D Golden Rice

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Qn	Answer	Qn	Answer
1	D	16	C
2	C	17	A
3	D	18	C
4	B	19	A
5	B	20	B
6	D	21	B
7	D	22	D
8	A	23	B
9	A	24	B
10	C	25	B
11	D	26	C
12	D	27	C
13	B	28	B
14	B	29	D
15	B	30	C